

Errata and minor updates

Physical Geodesy: Global Elastic Deformation of the Earth

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1. Page 236, from eq (4.40) to eq (4.43):

$$\begin{aligned} \Pi &> \left[1 - \frac{\alpha}{(N+1)N} \right] \cdots \left[1 - \frac{\alpha}{(i-3)(i-4)} \right] \left[1 - \frac{\alpha}{(i-1)(i-2)} \right] \\ &\times \left[1 - \frac{\alpha}{(i+1)i} \right] \left[1 - \frac{\alpha}{(i+3)(i+2)} \right] \cdots \left[1 - \frac{\alpha}{(i+l+1)(i+l)} \right] \cdots \end{aligned} \quad (4.40)$$

Take the logarithm on both sides,

$$\begin{aligned} \ln \Pi &> \ln \left[1 - \frac{\alpha}{(N+1)N} \right] + \cdots + \ln \left[1 - \frac{\alpha}{(i-3)(i-4)} \right] \\ &+ \ln \left[1 - \frac{\alpha}{(i-1)(i-2)} \right] + \ln \left[1 - \frac{\alpha}{(i+1)i} \right] \\ &+ \ln \left[1 - \frac{\alpha}{(i+3)(i+2)} \right] + \cdots + \ln \left[1 - \frac{\alpha}{(i+l+1)(i+l)} \right] + \cdots \end{aligned} \quad (4.41)$$

As $\lim_{l \rightarrow \infty} \{\alpha / [(i+l+1)(i+l)]\} = 0$, we have

$$\lim_{l \rightarrow \infty} \left\{ \ln \left[1 - \frac{\alpha}{(i+l+1)(i+l)} \right] \middle/ \left[-\frac{\alpha}{(i+l+1)(i+l)} \right] \right\} = 1, \quad (4.42)$$

which can be considered as an alternative form of the formula $\lim_{x \rightarrow 0} [\ln(1+x)]/x = 1$. Hence, the infinite series at the right-hand side of (4.41) converges or diverges in the same way as the series

$$\begin{aligned} \frac{\alpha}{(N+1)N} + \cdots + \frac{\alpha}{(i-3)(i-4)} + \frac{\alpha}{(i-1)(i-2)} + \frac{\alpha}{(i+1)i} \\ + \frac{\alpha}{(i+3)(i+2)} + \cdots + \frac{\alpha}{(i+l+1)(i+l)} + \cdots, \end{aligned} \quad (4.43)$$

2. Page 280, left hand side of the second line of eqn (4.296):
Replace “ a_k ” by “ b_k ”
3. Page 281, in eqn (4.302):
Replace “ 2π ” by “ 4π ”

4. Page 283, line 1 below (4.314):

Change to “This formula holds for arbitrarily large N . Therefore, the positive term series” and change “ $\sum_{n=0}^N$ ” to “ $\sum_{n=0}^{\infty}$ ” in (4.315)